Template Document

Academic Support Agent

February 19, 2025

This is the title in the language of the thesis An subtitle in the language of the thesis
Detta r den svenska vers ttningen av titeln Detta r den svenska vers ttningen av undertiteln
Student Fake A. a@kth.se u100001 0000-0002-00001-1234 EECS
Student Fake B. b@kth.se u100002 0000-0002-00001-5678 ABE
Supervisor A. Busy sa@kth.se u100003 EECS Computer Science
Supervisor Another Busy sb@kth.se u100003 ABE Public Buildings
Maguire Jr. Gerald Q. maguire@kth.se u100004 EECS Computer Science
F retaget AB
February 19, 2025
TCOMK

Abstract

[inline]The first abstract should be in the language of the thesis. [inline, backgroundcolor=aqua]Abstract fungerar p svenska ocks .

[inline]Keep in mind that most of your potential readers are only going to read your title and abstract. This is why it is important that the abstract give them enough information that they can decide is this document relevant to them or not. Otherwise the likely default choice is to ignore the rest of your document.
A abstract should stand on its own, i.e., no citations, cross references to the body of the document, acronyms must be spelled out, ...
Write this early and revise as necessary. This will help keep you focused on what you are trying to do.

Write an abstractUse about 1/2 A4-page (250 and 350 words). with the following components:

- What is the topic area? (optional) Introduces the subject area for the project.
- Short problem statement
- Why was this problem worth a Master s thesis project? (i.e., why is the problem both significant and of a suitable degree of difficulty for a Master s thesis project? Why has no one else solved it yet?)
- How did you solve the problem? What was your method/insight?
- Results/Conclusions/Consequences/Impact: What are your key results/conclusions? What will others do based upon your results? What can be done now that you have finished - that could not be done before your thesis project was completed?[inline]The presentation of the results should be the main part of the abstract.

Nyckelord

5-6 nyckelordNyckelord som beskriver inneh llet i uppsatsrapporten

Keywords

5-6 keywords [inline]Choosing good keywords can help others to locate your paper, thesis, dissertation, ... and related work. Choose the most specific keyword from those used in your domain, see for example: ACM's Computing Classification System (2012) or (2014) IEEE Taxonomy.

Mechanics:

- The first letter of a keyword should be set with a capital letter and proper names should be capitalized as usual.
- Spell out acronyms and abbreviations.
- Avoid "stop words" - as they generally carry little or no information.
- List your keywords separated by commas (",").

Since you should have both English and Swedish keywords - you might think of ordering them in corresponding order (i.e., so that the nth word in each list correspond) - thus it would be easier to mechanically find matching keywords.

?abstractname?

[inline]All theses at KTH are required to have an abstract in both English and Swedish.
If you are writing your thesis in English, you can leave this until the final version. If you are writing your thesis in Swedish then this should be done first – and you should revise as necessary along the way.
If you are writing your thesis in English, then this section can be a summary targeted at a more general reader. However, if you are writing your thesis in Swedish, then the reverse is true – your abstract should be for your target audience, while an English summary can be written targeted at a more general audience.
This means that the English abstract and Swedish sammnfattning or Swedish abstract and English summary need not be literal translations of each other.

The abstract in the language used for the thesis should be the first abstract, while the Summary/Sammanfattning in the other language can follow.

Exchange students many want to include one or more abstracts in the language(s) used in their home institutions to avoid the neeed to write another thesis when returing to their home institution.

Nyckelord

5-6 nyckelordNyckelord som beskriver inneh llet i uppsatsrapporten

[inline]Use the relevant language for abstracts for your home university.
Note that you may need to augment the set of lanaguage used in polyglossia or babel. The following languages represent the languages that have been used in theses at KTH in 2018-2019, except for one in Chinese.

?abstractname?

R sum en fran ais

Mots cl s

5-6 mots-cl s

?abstractname?

R sum en espagnol

Palabras claves

5-6 Palabras claves

?abstractname?

Sammendrag p norsk

N kkelord

5-6 n kkelord

?abstractname?

Abstract in Deutsch

Schl sselw rter

5-6 Schl sselw rter

?abstractname?

Abstrakt p dansk

S geord

5-6 S geord

?abstractname?

Samenvatting in het Nederlands

Trefwoorden

5-6 trefwoorden

?abstractname?

Eesti keeles kokkuv te

M rks nad

5-6 M rks nad

Acknowledgments

[inline]It is nice to acknowledge the people that have helped you. It is also necessary to acknowledge any special permissions that you have gotten – for example getting permission from the copyright owner to reproduce a figure. In this case you should acknowledge them and this permission here and in the figure s caption.
Note: If you do not have the copyright owner s permission, then you cannot use any copyrighted figures/tables/... .
[inline, backgroundcolor=aqua]I detta kapitel kan du e v n mna n got om din bakgrund om det p verkar rapporten p n got s tt. Har du t ex inte m jlighet att skriva perfekt svenska f r att du r nyanl nd till landet kan det vara p sin plats att n mna detta h r. OBS, detta f r dock inte vara en urs kt f r att l mna in en rapport med underm ligt spr k, grammatik och stavning (t ex f r fel som en automatisk stavningskontroll och grammatikkontroll kan uppt cka inte f rekomma)

En dualism som m ste hanteras i hela rapporten och projektet
I would like to thank xxxx for having yyyy.

Listings

1 Hello world in C code 11
2 Using a python program to access the KTH API to get all of the programs at KTH 11

If you have listings in your thesis.

[type=, title=List of acronyms and abbreviations] [inline]The list of acronyms and abbreviations should be in alphabetical order based on the spelling of the acronym or abbreviation.

Introduction [inline, backgroundcolor=aqua]svensk: Introduktion

[inline, backgroundcolor=aqua]Ofta kommer problemet och problem garen fr n industrin d r man nskar en specifik l sning p ett specifikt problem. Detta r ofta f r smalt definierat och ger ofta en f r smal l sning f r att resultatet skall vara intressant ur ett mer allm nt ingenj rsperspektiv och med nya erfarenheter som resultat. Fundera tillsammans med projektets intressenter (student, problem gare och akademi) hur man skulle kunna anv nda det aktuella problemet/f rslaget f r att unders ka n gon ingenj rsaspekt och vars resultat kan ge ny eller kompletterande erfarenhet till ingenj rssamfundet och vetenskapen.

Examensarbetet handlar d om att ta fram denna nya erfarenhet och p k pet l ser man en del eller hela delen av det ursprungliga problemet.

Erfarenheten kommer ur en fr gest llning som man i examensarbetet f rs ker besvara med tidigare och andras erfarenhet, egna eller modifierade metoder som ger ett resultat vilket kan anv ndas f r att diskutera ett svar p unders kningsfr gan.

Detta stycke skall allts , f rutom det ursprungliga smala problemet, inneh lla vad som skall unders kas f r att skapa ny ingenj rserfarenhet och/eller vetenskap.

ThisThe first paragraph after a heading is not indented, all of the subsequent paragraphs have their first line indented. chapter describes the specific problem that this thesis addresses, the context of the problem, the goals of this thesis project, and outlines the structure of the thesis.

[inline]Give a general introduction to the area. (Remember to use appropriate references in this and all other sections.)

We use the *bibtex* package to handle our references. We therefore use the command [**?**]. For example, Farshin, et al. described how to improve LLC cache performance in [**?**] in the context of links running at 200Gbps.

In this thesis we will examine the use of LAN. In this thesis we will assume that LAN to include WLAN, such as WiFi.

1 Background

[inline, backgroundcolor=aqua]svensk: Bakgrund Present the background for the area. Set the context for your project – so that your reader can understand both your project and this thesis. (Give detailed background information in Chapter 2 - together with related work.) Sometimes it is useful to insert a system diagram here so that the reader knows what are the different elements and their relationship to each other. This also introduces the

names/terms/... that you are going to use throughout your thesis (be consistent). This figure will also help you later delimit what you are going to do and what others have done or will do.

As one can find in RFC 1235[**?**] multicast is useful for xxxx .

2 Problem

[inline, backgroundcolor=aqua]svensk: Problemdefinition elle Fr gest llning
Lyft fram det ursprungliga problemet om det finns n got och definiera d refter den ingenj rsm ssiga erfarenheten eller/och vetenskapen som kan komma ur projektet.

Longer problem statement
If possible, end this section with a question as a problem statement.

2.1 Original problem and definition

[inline, backgroundcolor=aqua]Ursprungligt problem och definition Some text

2.2 Scientific and engineering issues

[inline, backgroundcolor=aqua]Vetenskaplig och ingenj rsm ssig fr gest llning some text

3 Purpose

[inline, backgroundcolor=aqua]Syfte State the purpose of your thesis and the purpose of your degree project.

Describe who benefits and how they benefit if you achieve your goals. Include anticipated ethical, sustainability, social issues, etc. related to your project. (Return to these in your reflections in Section 28.)

[inline, backgroundcolor=aqua]Skilj p syfte och m l! Syfte r att f r ndra n got till det b ttre. I examensarbetet finns ofta tv aspekter p detta. Dels vill problem garen (f retaget) f sitt problem l st till det b ttre men akademin och ingenj rssamfundet vill ocks f nya erfarenheter och vetskap. Beskriv ett syfte som tillfredst ller b da dessa aspekter. Det finns ven ett syfte till som kan vara v rt att beakta och det r att du som student skall ta examen och att du m ste bevisa, i ditt examensarbete, att du uppfyller examensm len. Dessa m l sammanfaller med kursm len f r examensarbetskursen.

4 Goals

[inline, backgroundcolor=aqua]M l [inline, backgroundcolor=aqua]Skilj p syfte och m l. Syftet r att stakomma en f r ndring i n got. M len r vad som konkret skall g ras f r att om m jligt uppn den nskade f r ndringen (syfte).

State the goal/goals of this degree project.

The goal of this project is XXX. This has been divided into the following three sub-goals:

1. Subgoal 1 [inline, backgroundcolor=aqua]f r att tillfredsst lla problem garen – industrin?
2. Subgoal 2[inline, backgroundcolor=aqua]f r att tillfredsst lla ingenj rssamfundet och vetenskapen – akademin)
3. Subgoal 3[inline, backgroundcolor=aqua]eventuellt, f r att uppfylla kursm len – du som student

In addition to presenting the goal(s), you might also state what the deliverables and results of the project are.

5 Research Methodology

[inline, backgroundcolor=aqua]Unders kningsmetod [inline, backgroundcolor=aqua]H r anger du vilken vilken vergripande unders kningsstrategi eller metod du skall anv nda f r att f rs ka besvara den akademiska fr gest llning och samtidigt l sa det e v ursprungliga problemet. Ofta kan man anv nda l sandet av ursprungsproblemet som en fallstudie kring en akademisk fr gest llning. Du unders ker n gon intressant fr ga i skarpt l ge och samlar resultat och erfarenhet ur detta.
T nk p att f retaget ibland m ste st tillbaka i sin nskan och f rv ntan p projektets resultat till f rm n f r ny eller kompletterande ingenj rserfarenhet och vetenskap (ditt examensarbete). Det r du som student som best mmer och l ser f rdelningen mellan dessa tv intressen men se till att alla r informerade. Introduce your choice of methodology/methodologies and method/methods – and the reason why you chose them. Contrast them with and explain

why you did not choose other methodologies or methods. (The details of the actual methodology and method you have chosen will be given in Chapter 11. Note that in Chapter 11, the focus could be research strategies, data collection, data analysis, and quality assurance.)
In this section you should present your philosophical assumption(s), research method(s), and research approach(es).

6 Delimitations

[inline, backgroundcolor=aqua]Avgr nsningar Describe the boundary/limits of your thesis project and what you are explicitly not going to do. This will help you bound your efforts – as you have clearly defined what is out of the scope of this thesis project. Explain the delimitations. These are all the things that could affect the study if they were examined and included in the degree project.

7 Structure of the thesis

[inline, backgroundcolor=aqua] Rapportens disposition Chapter 7 presents relevant background information about xxx. Chapter 11 presents the methodology and method used to solve the problem. ...

Background [inline, backgroundcolor=aqua]Bakgrund [inline]When you do your literature study, you should have a nearly complete Chapters 1 and 2.
You may also find it convenient to introduce the future work section into your report early – so that you can put things that you think about but decide not to do now into this section.
Note that later you can move things between this future work section and what you have done as you may change your mind about what to do now versus what to put off to future work.

This chapter provides basic background information about xxx. Additionally, this chapter describes xxx. The chapter also describes related work xxxx.

What does a reader (another x student – where x is your study line) need to know to understand your report? What have others already done? (This is the related work .) Explain what and how prior work / prior research will be applied on or used in the degree project /work (described in this thesis). Explain why and what is not used in the degree project and give valid reasons for rejecting the work/research.

[inline, backgroundcolor=aqua]Vilken viktig litteratur och (forsknings-)artiklar har du studerat inom omr det (litteraturstudie)?

8 Major background area 1

[inline, backgroundcolor=aqua]Viktigt bakgrundsomr de 1 There are xxx characteristics that distinguish yyy from other information and communication technology (ICT) system, as shown in Figure 1. Table 1 summarizes these characteristics.

[inline, backgroundcolor=aqua]Massor av st rnor (Inspirerad av figur x.y p sidan z i [xxx])

Table 1: xxx characteristics

Characteristics	**Description**
α	β
1	1110.1
2	10.1
3	23.113231

[inline, backgroundcolor=aqua]Egenskaper [inline, backgroundcolor=aqua] Beskrivning

8.1 Subarea 1.1

Entangled states are an important part of quantum cryptography, but also relevant in other domains. This concept might be relevant for neutrinos, see for example [**?**].

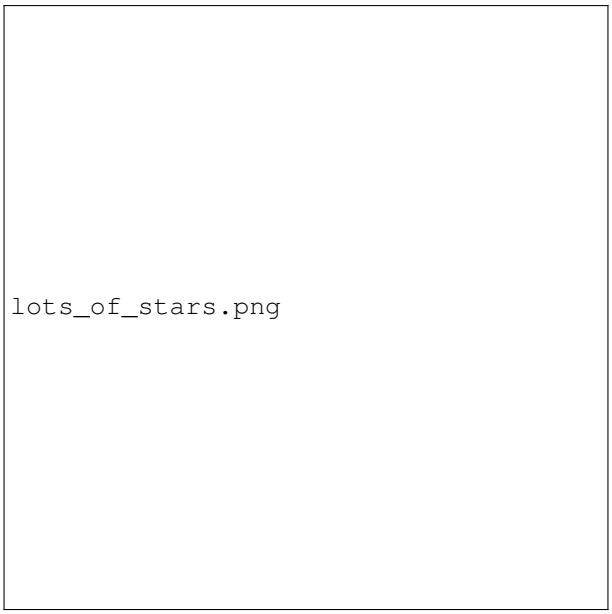


Figure 1: Lots of stars (Inspired by Figure x.y on page z of [xxx])

8.2 Subarea 1.1.2

Computational methods are increasingly used as a third method of carrying out scientific investigations. For example, computational experiments were used to find the amount of wear in a polyethylene liner of a hip prosthesis in [**?**]. ...

8.3 Subarea 1.1.2

Using the nearest data center may improve performance, see [**?**]

8.4 Link layer Encapsulation

See Figure **??** which uses the bytefield LaTeX package.

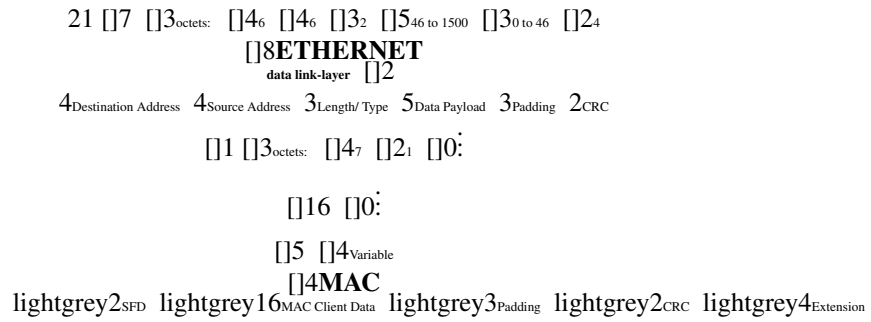


Figure 2: Ethernet data link layer protocol encapsulated into a IEEE 802.3 MAC packet

8.5 IP packet headers

The data link layer will receive a packet from the IP layer. The layout of an IPv4 packet is shown in Figure **??**. This should be contrasted with the IPv6 header shown in Figure **??**.

...

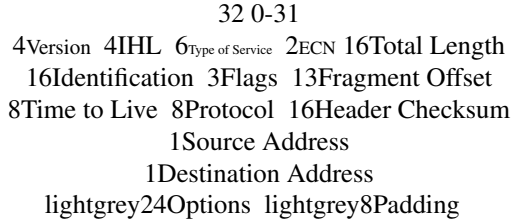


Figure 3: IPv4 datagram header. Light grey coloured fields are optional.

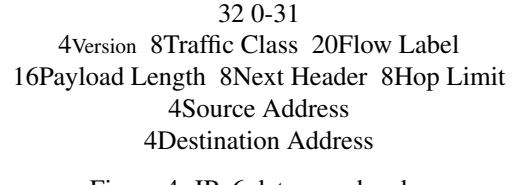


Figure 4: IPv6 datagram header

9 Major background area 2

[inline, backgroundcolor=aqua]Viktigt bakgrundsomr de 2 ...

10 Related work area

[inline, backgroundcolor=aqua]Relaterande arbeten

10.1 Major related work 1

[inline, backgroundcolor=aqua]Relaterande arbeten 1 Carrier clouds have been suggested as a way to reduce the delay between the users and the cloud server that is providing them with content. However, there is a question of how to find the available resources in such a carrier cloud. One approach has been to disseminate resource information using an extension to OSPF-TE, see Roozbeh, Sefidcon, and Maguire [**?**].

10.2 Major related work

[inline, backgroundcolor=aqua]Relaterande arbeten

10.3 Minor related work 1

[inline, backgroundcolor=aqua]Mindre relaterat arbete 1

...

10.4 Minor related work n

[inline, backgroundcolor=aqua]Mindre relaterat arbete n

11 Summary

[inline, backgroundcolor=aqua]Sammanfattning [inline, backgroundcolor=aqua]Det r trevligt att f detta kapitel avslutas med en sammanfattning. Till exempel kan du inkludera en tabell som sammanfattar andras id er och f rdelar och nackdelar med varje - s som senare kan du j mf ra din l sning till var och en av dessa. Detta kommer ocks att hj lpa dig att definiera de variabler som du kommer att anv nda f r din utv rdering.

[inline]It is nice to have this chapter conclude with a summary. For example, you can include a table that summarizes other people's ideas and benefits and drawbacks with each - so as later you can compare your solution to each of them. This will also help you define the variables that you will use for your evaluation.

Method or Methods [inline, backgroundcolor=aqua]Metod eller Metodval [inline]This chapter is about Engineering-related content, Methodologies and Methods. Use a self-explaining title.
The contents and structure of this chapter will change with your choice of methodology and methods.

Describe the engineering-related contents (preferably with models) and the research methodology and methods that are used in the degree project.

Give a theoretical description of the scientific or engineering methodology are you going to use and why have you chosen this method. What other methods did you consider and why did you reject them.

In this chapter, you describe what engineering-related and scientific skills you are going to apply, such as modeling, analyzing, developing, and evaluating engineering-related and scientific content. The choice of these methods should be appropriate for the problem . Additionally, you should be consciousness of aspects relating to society and ethics (if applicable). The choices should also reflect your goals and what you (or someone else) should be able to do as a result of your solution - which could not be done well before you started.

The purpose of this chapter is to provide an overview of the research method used in this thesis. Section 12 describes the research process. Section 13 details the research paradigm. Section 14 focuses on the data collection techniques used for this research. Section 15 describes the experimental design. Section 16 explains the techniques used to evaluate the reliability and validity of the data collected. Section 17 describes the method used for the data analysis. Finally, Section 18 describes the framework selected to evaluate xxx.

[inline, backgroundcolor=aqua]Vilka vetenskapliga eller ingenj rsmetodik ska du anv nda och varf r har du valt den h r metoden. Vilka andra metoder gjorde du verv ga och varf r du avvisar dem. Vad r dina m l? (Vad ska du kunna g ra som ett resultat av din l sning - vilken inte kan g ras i god tid innan du b rjade) Vad du ska g ra? Hur? Varf r? Till exempel, om du har implementerat en artefakt vad gjorde du och varf r? Hur kommer ditt utv rdera den. Syftet med detta kapitel r att ge en versikt ver forsknings metod som anv nds i denna avhandling. Avsnitt 12 beskriver forskningsprocessen. Avsnitt 13 detaljer forskningen paradigm. Avsnitt 14 fokuserar p datainsamling tekniker som anv nds f r denna forskning. Avsnitt 15 beskriver experimentell design. Avsnitt 16 f rklarar de tekniker som anv nds f r att utv rdera tillf rlitligheten och giltigheten av de insamlade uppgifterna. Avsnitt 17 beskriver den metod som anv nds f r dataanalysen. Slutligen, Avsnitt 18 beskriver ramverket valts f r att utv rdera xxx.

Ofta kan man koppla ett antal f ljdfr gor till unders kningsfr gan och probleml sningen t ex

- Vilken process skall anv ndas f r konstruktion av l sningen och vilken process skall kopplas till denna f r att svara p unders kningsfr gan?
- Hur och vilket resultat (storheter) skall presenteras b de f r att redovisa svar p unders kningsfr gan (resultatkapitlet i denna rapport) och redovisa resultat av probleml sningen (prototypen, ofta dokument som bilagor men vilka dokument och varf r?).
- Vilken teori/teknik skall v ljas och anv ndas b de f r unders kningen (taxonomi, matematik, grafer, storheter mm) och probleml sning (UML, UseCases, Java mm) och varf r?
- Vad beh ver du som student leverera f r att uppn h g kvaliet (minimikrav) eller mycket h g kvalitet p examensarbetet? Fr gorna kopplar till de f ljande underkapitlen.[inline, backgroundcolor=aqua]Resonemanget bygger p att studenter p hing-programmet ofta skall konstruera n got t problem garen och att man till detta m ste koppla en intressant ingenj rsfr ga. Det finns hela tiden en dualism mellan dessa aspekter i exjobbet.

12 Research Process

[inline, backgroundcolor=aqua]Unders kningsrocess och utvecklingsprocess Figure 5 shows the steps conducted in order to carry out this research. Figur 5 visar de steg som utf rs f r att genomf ra

Beskriv, g rna med ett aktivitetsdiagram (UML?), din unders kningsprocess och utvecklingsprocess. Du m ste koppla ihop det akademiska intresset (unders kningsprocess) med ursprungsproblemet (utvecklingsprocess) denna forskning. [inline, backgroundcolor=aqua]Aktivitetsdiagram fr n t ex UML-standard

[inline, backgroundcolor=aqua]Forskningsprocessen

13 Research Paradigm

[inline, backgroundcolor=aqua]Unders kningsparadigm Exempelvis
Positivistisk (vad/hur fungerar det?) kvalitativ fallstudie med en deduktivt (f rbest md) vald ansats och ett induktivt(efterhand uppst r dataomr den och data) insamlade av data och erfarenheter.



Figure 5: Research Process

14 Data Collection

[inline]This should also show that you are aware of the social and ethical concerns that might be relevant to your data collection method.) [inline, backgroundcolor=aqua]Datainsamling (Detta b r ocks visa att du r medveten om de sociala och etiska fr gor som kan vara relevanta f r dina data insamlingsmetod.)

14.1 Sampling

[inline, backgroundcolor=aqua]Stickprovsunders kning

14.2 Sample Size

[inline, backgroundcolor=aqua]Provstorleken

14.3 Target Population

[inline, backgroundcolor=aqua]M lgruppen

15 Experimental design/Planned Measurements

[inline, backgroundcolor=aqua]Experimentdesign/M tuppst llning

15.1 Test environment/test bed/model

[inline]Describe everything that someone else would need to reproduce your test environment/test bed/model/... . [inline, backgroundcolor=aqua]Testmilj /testb dd/modell Beskriv allt att n gon annan skulle beh va terskapa din testmilj / testb dd / modell / ...

15.2 Hardware/Software to be used

[inline, backgroundcolor=aqua]H rdvara / programvara som ska anv ndas

16 Assessing reliability and validity of the data collected

[inline, backgroundcolor=aqua]Bed mning av validitet och reliabilitet hos anv nda metoder och insamlade data

16.1 Validity of method

[inline]How will you know if your results are valid? [inline, backgroundcolor=aqua]Giltigheten av metoder Har dina metoder ge dig de r tta svaren och l sning? Var metoderna korrekt?

16.2 Reliability of method

[inline]How will you know if your results are reliable? [inline, backgroundcolor=aqua]Tillf rlitlighet av metoder Hur bra r dina metoder, finns det b ttre metoder? Hur kan du f rb ttra dem?

16.3 Data validity

[inline, backgroundcolor=aqua]Giltigheten av uppgifter Hur vet du om dina resultat r giltiga? Har ditt resultat m ta r tta?

16.4 Reliability of data

[inline, backgroundcolor=aqua]Tillf rlitlighet av data Hur vet du om dina resultat r tillf rlitliga? Hur bra r dina resultat?

17 Planned Data Analysis

[inline, backgroundcolor=aqua]Metod f r analys av data

17.1 Data Analysis Technique

[inline, backgroundcolor=aqua]Dataanalys Teknik

17.2 Software Tools

[inline, backgroundcolor=aqua]Mjukvaruverktyg

18 Evaluation framework

[inline, backgroundcolor=aqua]Utv rdering och ramverk Metod f r utv rdering, j mf relse mm. Kopplar till kapitel 21.1.

19 System documentation

[inline]If this is going to be a complete document consider putting it in as an appendix, then just put the highlights here. [inline, backgroundcolor=aqua]Systemdokumentation Med vilka dokument och hur skall en konstruerad prototyp dokumenteras? Detta blir ofta bilagor till rapporten och det som problem garen till det ursprungliga problemet (industrin) ofta vill ha.
Bland dessa bilagor terfinns ofta, och enligt n gon angiven standard, kravdokument, arkitekturdokument, designdokumnet, implementationsdokument, driftsdokument, testprotokoll mm.

What you did[inline]Choose your own chapter title to describe this [inline, backgroundcolor=aqua][Vad gjorde du? Hur gick det till? – V lj l mplig rubrik (Genomf rande , Konstruktion , Utveckling eller annat]

[inline]What have you done? How did you do it? What design decisions did you make? How did what you did help you to meet your goals? Vad du har gjort? Hur gjorde du det? Vad designen beslut gjorde du? Hur kom det du hj lpte dig att uppn dina m l?

20 Hardware/Software design .../Model/Simulation model & parameters/...

[inline, backgroundcolor=aqua]H rdvara / Mjukvarudesign ... / modell / Simuleringsmodell och parametrar / ...

Figure 6 shows a simple icon for a home page. The time to access this page when served will be quantified in a series of experiments. The configurations that have been tested in the test bed are listed in Table 2.

Figur 6 visar en enkel ikon f r en hemsida. Tiden f r att f tillg ng till den h r sidan n r serveras kommer att kvantifieras i en serie experiment. De konfigurationer som har testats i provb nk listas ini tabell 2.

Vad du har gjort? Hur gjorde du det? Vad designen beslut gjorde du?



Figure 6: Homepage icon

Table 2: Configurations tested

Configuration	Description
1	Simple test with one server
2	Simple test with one server

[inline, backgroundcolor=aqua]Konfigurationer testade

21 Implementation .../Modeling/Simulation/...

[inline, backgroundcolor=aqua]Implementering ... / modellering / simulering / ...

21.1 Some examples of coding

Listing 1 shows an example of a simple program written in C code.

Listing 1: Hello world in C code

```
int main() {
printf("hello, world");
return 0;
}
```

In contrast, Listing 2 is an example of code in Python to get a list of all of the programs at KTH.

Listing 2: Using a python program to access the KTH API to get all of the programs at KTH

```
KOPPSbaseUrl = 'https://www.kth.se'

def v1_get_programmes():
    global Verbose_Flag
    #
    # Use the KOPPS API to get the data
    # note that this returns XML
    url = "{0}/api/kopps/v1/programme".format(KOPPSbaseUrl)
    if Verbose_Flag:
        print("url: " + url)
    #
    r = requests.get(url)
    if Verbose_Flag:
        print("result of getting v1 programme: {}".format(r.text))
    #
```

```
    if r.status_code == requests.codes.ok:
        return r.text           # simply return the XML
    #
    return None
```

Results and Analysis [inline, backgroundcolor=aqua]svensk: Resultat och Analys [inline] Sometimes this is split into two chapters.

Keep in mind: How you are going to evaluate what you have done? What are your metrics?
Analysis of your data and proposed solution
Does this meet the goals which you had when you started?

In this chapter, we present the results and discuss them.

I detta kapitel presenterar vi resultatet och diskutera dem. [inline, backgroundcolor=aqua] Ibland delas detta upp i tv kapitel.
Hur du ska utv rdera vad du har gjort? Vad r din statistik?
Analys av data och f reslagen l sning
Inneb r detta att uppn de m l som du hade n r du b rjade?

22 Major results

[inline, backgroundcolor=aqua]Huvudsakliga resultat

Some statistics of the delay measurements are shown in Table 3. The delay has been computed from the time the GET request is received until the response is sent.

Lite statistik av m tningarna f rdr jnings visas i Tabell 3. F rseningen har ber knats fr n den tidpunkt d beg ran GET tas emot fram till svaret skickas.

Table 3: Delay measurement statistics

Configuration	Average delay (ns)	Median delay (ns)
1	467.35	450.10
2	1687.5	901.23

[inline, backgroundcolor=aqua]F rdr j m tstatistik [inline, backgroundcolor=aqua]Konfiguration | Genomsnittlig f rdr jning (ns) | Median f rdr jning (ns)

Figure 7 shows and example of the performance as measured in the experiments.

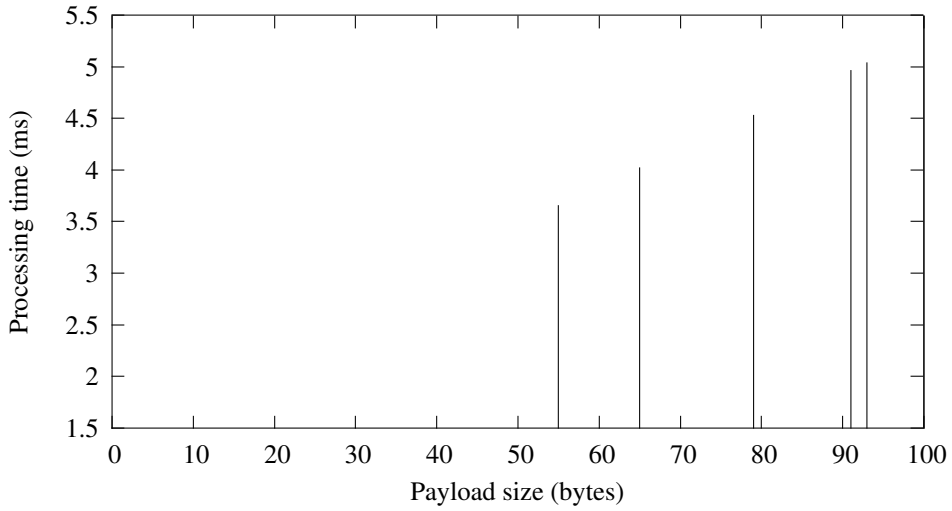


Figure 7: Processing time vs. payload length

Given these measurements, we can calculate our processing bit rate as the inverse of the time it takes to process an additional byte divided by 8 bits per byte:

$$bitrate = \frac{1}{\frac{time_{byte}}{8}} = 20.03 \quad kb/s$$

23 Reliability Analysis

[inline, backgroundcolor=aqua]Analys av reabilitet Reabilitet i metod och data

24 Validity Analysis

[inline, backgroundcolor=aqua]Analys av validitet Validitet i metod och data

Discussion[inline]This can be a separate chapter or a section in the previous chapter. [inline, backgroundcolor=aqua]Diskussion F rb ttringsf rslag?

Conclusions and Future work [inline, backgroundcolor=aqua]Slutsats och framtida arbete Add text to introduce the subsections of this chapter.

25 Conclusions

[inline]Describe the conclusions (reflect on the whole introduction given in Chapter 1). [inline, backgroundcolor=aqua]Slutsatser

Discuss the positive effects and the drawbacks.
Describe the evaluation of the results of the degree project.
Did you meet your goals?
What insights have you gained?
What suggestions can you give to others working in this area?
If you had it to do again, what would you have done differently?

Tr ffade du dina m l? Vilka insikter har du f tt? Vilka f rslag kan du ge till andra som arbetar inom detta omr de? Om du hade att g ra igen, vad skulle du ha gjort annorlunda?

26 Limitations

[inline]What did you find that limited your efforts? What are the limitations of your results? [inline, backgroundcolor=aqua]Begr nsande faktorer Vad gjorde du som begr nsade dina anstr ngningar? Vilka r begr nsningarna i dina resultat?

27 Future work

[inline]Describe valid future work that you or someone else could or should do.
Consider: What you have left undone? What are the next obvious things to be done? What hints can you give to the next person who is going to follow up on your work? [inline, backgroundcolor=aqua]Vad du har kvar ogjort?
Vad r n sta sj lvklara saker som ska g ras?
Vad tips kan du ge till n sta person som kommer att f lja upp p ditt arbete?

Due to the breadth of the problem, only some of the initial goals have been met. In these section we will focus on some of the remaining issues that should be addressed in future work. ...

27.1 What has been left undone?

The prototype does not address the third requirment, i.e., a yearly unavailability of less than 3 minutes, this remains an open problem. ...

27.1.1 Cost analysis

The current prototype works, but the performance from a cost perspective makes this an impractical solution. Future work must reduce the cost of this solution, to do so a cost analysis needs to first be done. ...

27.1.2 Security

A future research effort is needed to address the security holes that results from using a self-signed certificate. Page filling text mass. Page filling text mass. ...

27.2 Next obvious things to be done

In particular, the author of this thesis wishes to point out xxxxxx remains as a problem to be solved. Solving this problem is the next thing that should be done. ...

28 Reflections

[inline]What are the relevant economic, social, environmental, and ethical aspects of your work? [inline, backgroundcolor=aqua]Reflektioner [inline, backgroundcolor=aqua]Vilka r de relevanta ekonomiska, sociala, milj m ssiga och etiska aspekter av ditt arbete?

One of the most important results is the reduction in the amount of energy required to process each packet while at the same time reducing the time required to process each packet.

The thesis contributes to the UN SDG numbers 1 and 9 by xxxx.

[inline]In the references, let Zotero or other tool fill this in for you. I suggest an extended version of the IEEE style, to include URLs, DOIs, ISBNs, etc., to make it easier for your reader to find them. This will make life easier for your opponents and examiner.

IEEE Editorial Style Manual: `https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/style_references_manual.pdf` [inline, backgroundcolor=aqua]L t Zotero eller annat verktyg fylla i det h r f r dig. Jag f resl r en ut kad version av IEEE stil - att inkludera webbadresser, DOI, ISBN etc. - f r att g ra det l ttare f r l saren att hitta dem. Detta kommer att g ra livet l ttare f r dina motst ndare och examinator.

Something Extra [inline, backgroundcolor=aqua]svensk: Extra Material som Bilaga

For DIVA

pg:lastPageofPrefacepg:lastPageofMainmatter